COURSE TITLE	**CLOUD COMPUTING**
COURSE CODE	**01CT1611**
COURSE CREDITS	**4**

Objective:

1 This course is intended to analyse the basics of cloud computing, and make aware students with diversified technologies working for cloud architecture. Course will be focusing on architecture, service models, privacy & security in the cloud.

Course Outcomes: After completion of this course, student will be able to:

1 Understand the architecture and infrastructure of cloud computing, including SaaS, PaaS, IaaS, public cloud, private cloud, hybrid cloud, various management and other distinguished services of AWS.

2 Apply the fundamental concepts in datacenters to understand the trade-offs in power, efficiency and cost by the Load balancing approach and instances.

3 Illustrate the fundamental concepts of cloud storage and demonstrate their use in storage systems such as Amazon S3 and Database.

4 Analyze various cloud Service models and apply them to solve problems on the cloud.

5 Deploy applications over commercial cloud computing infrastructures such as AWS

Pre-requisite of course: Operating System and Computer Networks

Teaching and Examination Scheme

Theory Hours	Tutorial Hours	Practical Hours	ESE	IA	CSE	Viva	Term Work
3	0	2	50	30	20	25	25

Contents : Unit	Topics	Contact Hours
1	**Introduction of Cloud & Amazon Web Service** Introduction of cloud computing, how it works Types of cloud, what is Virtualization, Advantages of Cloud, AWS history, Dashboard, AWS Overview, Architecture	5
2	**Cloud Service Models** Software as a Service (SaaS), Infrastructure As a Services (IaaS), Platform As a service (PaaS)	4
3	**Identity & Access Management** IAM Overview and Policies, IAM Users, Groups, Access Key, Secret Access Key, MFA, Report	5

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Contents : Unit	Topics	Contact Hours
4	**Elastic Cloud Computing (EC2)** Amazon EC2 Overview, Elastic Block Storage (EBS), Amazon Machine Image (AMI), Instance Purchasing Options, Introduction to EC2 Instance Types Security Group Elastic, Public & private IP Overview, Amazon EBS & Snapshot, AWS CLI, Bootstrap Script, Elastic Load Balancing (ELB), Auto Scaling	5
5	**Virtual Private Cloud (VPC)** Amazon Virtual Private Cloud (VPC), Amazon VPC and Subnets, Route Table, Internet Gateway	6
6	**Amazon Simple Storage Service (S3)** Simple Storage Service (S3), S3 Object Storage and Buckets, Security on bucket, Logging & event, Glacier, Versioning & Lifecycle Policy, Cross region replication	6
7	**Route 53** DNS Records, Website Hosting, Health Check	3
8	**Databases** Relation Database System, DB engine & Instance details, Security, Parameter group, Monitoring Resourcing, DynamoDB, Elasticache	5
9	**CloudWatch & Monitoring** Cloud Watch, Matrices, Alarm & notification	3
Total Hours		**42**

Suggested List of Experiments:

Contents : Unit	Topics	Contact Hours
1	**Experiment-1** Setup Windows Instance in AWS Cloud using EC2 service and install IIS	2
2	**Experiment-2** Setup Linux Instance in AWS Cloud using EC2 service and install Apache(httpd)	2
3	**Experiment-3** Add and modify features in Existing Windows Instance	2
4	**Experiment-4** Build Your Virtual Private Cloud (VPC) and Launch a Web Server	2
5	**Experiment-5** Working with Amazon Elastic Block Store (EBS)	2
6	**Experiment-6** Introduction to AWS Identity and Access Management (IAM)	2
7	**Experiment-7** Deploy a Web Application on AWS	2
8	**Experiment-8** Using Auto Scaling with AWS Lambda and Lifecycle Hooks	2

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Suggested List of Experiments:

Contents : Unit	Topics	Contact Hours
9	**Experiment-9** Implementing a Serverless Architecture with AWS Managed Service	2
10	**Experiment-10** Launching EC2 Spot Instances with Auto Scaling and Amazon CloudWatch	2
11	**Experiment-11** Create Static Website Using S3	2
12	**Experiment-12** Crease S3 Bucket and Upload and Download file	2
13	**Experiment-13** Create Lifecycle Policy using S3	2
14	**Experiment-14** Setup Versioning & Configure Cross Region Replication in S3	2
15	**Experiment-15** Create Load Balancer using EC2 Service	2
16	**Experiment-16** RDS : Create Database & Connect	2
17	**Experiment-17** Create Autoscaling Group and Configure Multi Factor Authentication	2
Total Hours		**34**

Textbook :

1 Cloud Computing for Dummies, Judith Hurwitz, R Bloor, M.Kanfman, F.Halper, Wiley India Edition, 2012

2 Cloud Computing: Principles and Paradigms, Rajkumar Buyya, James Broberg, Andrzej M. Goscinski, Wiley, 2011

References:

1 Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance, Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance, Tim Mather, SubraKumara swamy, Shahed Latif, O'ReillyMedia Inc, 2009

2 IT Virtualization Best Practices: A Lean, Green Virtualized Data Center Approach, IT Virtualization Best Practices: A Lean, Green Virtualized Data Center Approach, Mickey Iqbal, MC Press , 2010

3 Mobile Clouds: Exploiting Distributed Resources in Wireless, Mobile and Social Networks, Mobile Clouds: Exploiting Distributed Resources in Wireless, Mobile and Social Networks, Frank H. P. Fitzek, Marcos D. Katz, Wiley, 2014

Suggested Theory Distribution:

The suggested theory distribution as per Bloom's taxonomy is as follows. This distribution serves as guidelines for teachers and students to achieve effective teaching-learning process

Distribution of Theory for course delivery					
Remember / Knowledge	**Understand**	**Apply**	**Analyze**	**Evaluate**	**Higher order Thinking / Creative**
5.00	15.00	20.00	20.00	20.00	20.00

Instructional Method:

1. The internal evaluation will be done on the basis of continuous evaluation of students in the laboratory and class-room.
2. Practical examination will be conducted at the end of the semester for evaluation of performance of students in laboratory.
3. Students may use supplementary resources such as online videos, NPTEL videos, e-courses, Virtual Laboratory, etc.
4. The course delivery method will depend upon the requirement of content and need of the students. The teacher in addition to conventional teaching method (Chalk and Talk) may use any of the tools such as demonstration, role play, Quiz, brainstorming, Flipped class, Project based learning, Collaborative learning, MOOCs etc. for effective teaching.

Supplementary Resources:

1. https://onlinecourses.nptel.ac.in/noc17_cs23/preview
2. https://www.edx.org/micromasters/cloud-computing
3. https://aws.amazon.com/training/awsacademy/cloud-computing-architecture
4. https://www.coursera.org/specializations/cloud-computing

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